Deterioration states — methodology

Collections risk modelling · deterioration states · docs/deterioration_states/METHODOLOGY.md

How a credit-card account in buckets 4–7 fell into delinquency, and what it is doing now, read from its DPD bucket history only. One state per account, assigned by a short decision tree. MECE by construction, simple enough to explain to the risk team in one slide, and designed to stay out of the way of the other persona axes.

> **Status.** Designed and validated on calibrated **synthetic** data, then independently reviewed (24 findings — see REVIEW.md for each finding and what was done). Not yet run on the bank's book. Every number marked *(synthetic)* is a *method check*, not a finding about the portfolio. The real-data run is `run.py validate` on a file (now) or `snippets/dpd_states_validate.py` on Spark (production); expectations in §8.

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1. Where this sits among the persona axes

Axis	Question it answers	Inputs
Locatability	Can we reach them?	Reachability flag, in-country flag
Ability	Can they pay?	Score: salary, inflow, CASA balance
Payment credibility	Do they pay and keep their word?	Score: promises kept (0–2), money coming in last 3 months (0–2), pre-delinquency history clean (0–1)
Deterioration (this)	**How did they fall in, and are they stabilized now?**	**DPD bucket history only**

Design rules agreed with the business owner:

1. **MECE** — every account gets exactly one state; no "other" bucket for behaviour.
2. **Simplicity beats precision** — a rule a collector can repeat beats a better-fitting one nobody can explain. Thresholds are round numbers.
3. **Orthogonal where possible** — the states must not rebuild the credibility or ability score. Known overlaps are named (§7), not hidden.
4. **5–6 states, at most 8**, each with a different action for the risk team.

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2. Population and data conventions

- **Spine:** credit-card accounts whose **current** bucket is 4, 5, 6 or 7. Bucket 8+ is a separate category. The input file for sign-off is the CC 60–180 DPD segment (Aug 2026); accounts at bucket 3 in it are outside the spine and filtered out.
- **The string:** one digit per month, **leftmost = current month** (`54320002000` → today 5, last month 4, …). The **oldest (rightmost) digit is the first billed month**, so it is almost always 0 or 1. Exceptions: rare missing history, 180+ accounts without history (start at 9, outside the spine), and restructured cards/loans re-opened as a new account at their previous bucket.

- **Reliable window: January 2024 onwards.** Older DPD history is unreliable, so every string is cropped to the newest months from `RELIABLE_FROM` (2024-01) to `SNAPSHOT_MONTH` (2026-08) — **32 months** — or to the source's own fixed window if shorter (`SOURCE_WINDOW_MONTHS` ; the sign-off file now carries full history, so 32 applies). A string that fills the whole window may have older history and is treated as cropped — never EPD, never "history starts delinquent", and a straight roller is seasoned. Effects, all handled: - EPD is unaffected: an EPD has at most 3 clean months plus a short straight roll, so its string is under 12 digits and never cropped. As a safety net, a cropped string can never be EPD (its oldest digit is no longer the first billed month). - A spell that started before 2024 makes a cropped string start delinquent. That is normal history, not a data issue, so Q0 skips cropped strings; these accounts (0.16% of the synthetic spine) are classified normally — in practice mostly stabilized. - Seasoned (≥ 12 months) and the 12-month repeat window fit inside 32 months; earlier spells that ended before 2024 are invisible, which can slightly under-count repeat delinquents whose spell began in early 2024. *(Synthetic: 74% of strings cropped; option-B shares essentially unchanged.)*
- **Read it as text.** A numeric column drops digits (Excel keeps 15 significant digits; leading zeros vanish). The file reader refuses number-like values.
- **0 ≡ 1.** Bucket 1 (a few days late) is treated as normal.
- **Monthly movement `d` = bucket this month − bucket last month**, which reads as a payment:

d	Meaning	Months paid that month (= 1 − d)
+1	roll — paid nothing (or less than one month's due)	0
0 at bucket ≥ 2	**hold** — paid one month's due, stays in the same bucket	1
−k	paid k + 1 months, moved back k buckets	k + 1
back to 0/1	**cure** — common from 2–3, very rare from 4+	—

- **Spell:** a continuous run of months at bucket ≥ 2. The *current spell* is the one the account is in today.
- **Blip:** an earlier spell of 1–2 months that never passed bucket 2 (`0 0 2 0 0`). Blips do not count as earlier spells and do not end the clean history.
- **Data artefacts are cleaned before anything else, and counted:** - *Impossible one-month glitches.* A single 0/1 month between two months at 3+ (`4 0 4` : a cure cannot jump straight back to 4) is read as a hold; a single month at 3+ between two clean months (`0 3 0` : an account cannot jump from 0 to 3) is read as clean. Neither creates an earlier spell. - *Billing-cycle +2 jumps.* A true `3 4 5` reported as `3 3 5` or `3 5 5` is repaired. The hold *before* the jump is preferred as the artefact (it is already observed); the hold after is used only if that does not move a month observed at 4 back below 4; a hold at 0/1 is never an artefact. The repair is therefore causal: a state assigned today does not change when next month's digit arrives. Any other +2 is read as a roll.

3. Features used by the states

Feature	Definition	Fixed once past 4?
`left_censored`	the oldest month in the string is already at bucket 2+	yes
`months_before_spell`	months of history before the current spell started (blip months included)	yes
`n_blips`	earlier 1–2 month excursions that never passed bucket 2 (EPD allows one)	yes
`n_prior_episodes`	earlier real (non-blip) spells (EPD requires none)	yes

Feature	Definition	Fixed once past 4?
`gap_prior_peak3`	clean months between the end of the most recent earlier spell that reached bucket 3 and the start of this one (repeat delinquent: ≤ 12)	yes
`spell_pay_events`	months in the current spell with any payment (hold or move back); EPD requires 0	**no** — EPD ends when they pay, by design
`runup_len`	months from the last clean month to the first month at bucket 4 (3 = straight 2→3→4)	yes
`runup_pays`	months on the way to 4 in which 2+ months were paid (moved back)	yes
`last6_in_band`	of the last 6 months (incl. today), how many were within today's bucket ±1	**no** — current behaviour, by design
`months_since_4`, `chg_6m`	months since first reaching 4 in this spell; today's bucket minus the bucket 6 months ago (long-run clause)	**no** — current behaviour, by design

All other features (`runup_holds` , `post4_paid_rate` , `max_hold_run` , …) are diagnostics only. The current-behaviour inputs are confined to the stabilized question and the EPD "never paid" condition, so every other state is fixed once the account is past 4.

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4. The states

Questions are asked **in order**; the first "yes" assigns the state.

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Q0  History starts delinquent?  oldest month already at bucket 2+ (restructured re-open,
                                migrated or missing history)  -> history_starts_delinquent
Q1  Early payment default?  <= 3 months of history before this spell (at most one blip,
                           no earlier real spell) AND never paid since crossing bucket 2
                           (every month of the spell a roll — still falling)
                                                    -> early_payment_default
Q2  Stabilized?            >= 5 of the last 6 months within today's bucket ±1
                           OR (>= 6 months since first reaching 4 AND at most 1 bucket
                               higher than 6 months ago)      -> stabilized
Q4  Repeat delinquent?     [B, C] an earlier spell reached bucket 3, cured, and ended
                           <= 12 months before this spell started
                                                    -> repeat_delinquent
Q5  Straight roller?       run-up (last clean month -> first month at 4) <= 3 months
      yes -> straight_roller  [B, C: >= 12 months of history before the spell
                               -> straight_roller_seasoned, else
straight_roller_early_life]
      no  -> slow_roller      [C: paid 2+ months at once on the way down
                               -> slow_roller_erratic, else slow_roller_intermittent]
```

(Q3 "paying since bucket 4" was tested and rejected — §6.4.) Q0 is a data-quality guard, not a behaviour state: it stops truncated or restructured histories from being forced into EPD. It should be rare (E04); restructured accounts are a deferred decision.

Three options

Option	States	Use when
A — minimal (4)	EPD, stabilized, straight roller, slow roller	Strict orthogonality: prior-spell history is left entirely to credibility
B — balanced (6), recommended	EPD, stabilized, repeat delinquent, seasoned straight roller, early-life straight roller, slow roller	Default; the 5–6 target
C — detailed (7)	B with slow roller split into intermittent payer / erratic payer	The risk team has a different action for each and both are ≥ 4%

Early-life straight roller stays a state in B and C, but if real data puts it under 5% (E16) it is merged into straight roller and kept as an early-life flag (agreed).

What each state means and the action it points to

State (display name · `code`)	Reads as	Action flavour	Share *(synthetic)*
Early payment default (EPD) · `early_payment_default`	At most 3 months of history (one blip allowed), rolled straight in and has **never paid since crossing bucket 2** — still falling, so short-tenure by nature	Early-risk / first-party-fraud review, early legal; no restructuring	4.3% (bank's broader rule: 6%)
Stabilized · `stabilized`	Paying about one month's due each month for 5 of 6 months — or, 6+ months after first reaching 4, still within one bucket of where it was 6 months ago (pays what is due on average, unevenly)	**Offer restructuring to break the hardship**	7.8% (bank's rule alone: 6.9%; bank: 6%)
Repeat delinquent (re-default) · `repeat_delinquent`	Cured a spell that reached bucket 3 and fell back in within 12 months	Has shown they can catch up: firm promise-based plan; credibility decides the pressure	10.8%
Seasoned straight roller · `straight_roller_seasoned`	At least a year on book, then rolled straight to bucket 4 with no payment on the way down (some have paid since)	Life event. Read with the other axes: not in country → departed; no salary → income shock; salary still in → won't pay. Most likely to respond to a hardship route	57.4%
Early-life straight roller · `straight_roller_early_life`	4–11 months on book, then rolled straight to bucket 4	Same triage; also an underwriting signal for acquisition	4.7% — under the 5% floor in synthetic data; the real share decides (E16)
Slow roller (partial payer) · `slow_roller`	Paid something on the way down to bucket 4 but kept rolling	Over-indebted: longer-tenor restructure or settlement	15.1%

State (display name · `code`)	Reads as	Action flavour	Share *(synthetic)*
↳ **Intermittent payer** *(C)* · `slow_roller_intermittent`	Paid one month here and there on the way down, never caught up	Chronic insufficiency: restructure to what they actually pay	9.3%
↳ **Erratic payer** *(C)* · `slow_roller_erratic`	Caught up 2+ months at once on the way down, then rolled again	Lumpy / irregular income: plan aligned to inflow timing	5.7%
Not classified — history starts delinquent · `history_starts_delinquent`	Data quality: the string starts already delinquent	Fix the data / treat restructured accounts separately (deferred)	0% *(synthetic has none)*; expect < 1%

In option A the two straight-roller states are one: **Straight roller** · `straight_roller`.

What moves month to month. Stabilized is current behaviour (*synthetic:* 60% are still stabilized next month). EPD ends the month the account pays (it then becomes stabilized or a straight roller). Every other state describes how the account fell in and is fixed once it is past bucket 4 — it can only move into stabilized (*synthetic:* 99.7%+ keep their state or move into stabilized next month). Read "straight roller" as *rolled straight to bucket 4*, not *never paid since*: 32–36% of straight rollers have paid at least one month since reaching 4 *(review, synthetic)*.

4.1 Names

The names use established card-risk and collections terms, and follow two rules:

1. **Describe behaviour, not intent.** These labels reach collector screens, audit trails and possibly complaint files. They say what the DPD history shows ("never paid since crossing 2", "rolled straight to 4", "paid one month at a time"), never why ("fraud", "won't pay"), which the history cannot prove.
2. **Code names are stable; display names are free.** Code uses short identifiers (`early_payment_default`, `straight_roller_seasoned`, …). `DISPLAY_NAMES` and `REMARKS` in `dpd_states/states.py` carry the business labels, so renaming for the risk team never breaks the pipeline or old results. Every report starts with the names table.

Term	Where it comes from	Note
Early payment default (EPD)	Early-risk monitoring (FPD / SPD / TPD = first / second / third payment default; EPD = default early in the life of the account)	**Formerly "bust-out"** — the bank's internal name, kept in remarks. Classic *bust-out* fraud is different: a customer builds a good record and limit, maxes out (often with cash advances) and disappears. Its DPD history is clean-then-straight-roll, so it lands in **seasoned straight roller**; spotting it needs spend / utilisation spikes before delinquency — a fraud overlay, not this axis
Stabilized	The bank's own term; roll-rate "stay" behaviour	Industry has no single standard label for "pays enough to stay in the bucket"
Repeat delinquent / re-default	Collections recidivism; re-default rates after cure or restructure, usually within 12–24 months	
Straight roller	Core card-collections term: rolls through the buckets without a payment	Here: on the way down to bucket 4

Term	Where it comes from	Note
Seasoned / early-life	Seasoning and months on book (MOB); the first 12 months is the standard early-life window	
Slow roller (partial payer)	Roll-rate language: the counterpart of the straight roller	**Also called partial payer** — the more common collections term. The rejected Q3 state was renamed `paying_since_4` so "partial payer" means only this state
Intermittent / erratic payer	Collections payment-pattern labels	

5. Thresholds and the evidence behind them

Parameters marked **[business]** are agreed business definitions. They change only with the business owner — never to make a share or an expectation check come out right (a share-matching tune is circular: §6.14). Sweeps are reported in every run to show sensitivity, not as a menu.

Threshold (`parameter`)	Value	Why this value	What moving it does *(synthetic)*
EPD history (`epd_max_history` , `epd_max_blips`) [business]	**≤ 3 months, at most one blip**	A customer who never pays misses the first or second bill: 2–3 clean months at most (`0 0 2 3 4` , `0 0 0 2 3 4` , `0 2 0 2 3 4`)	At 6 or 12 months the state fills with early-life defaults (`0 0 0 0 0 0 2 3 4`), which now land in early-life straight roller
EPD "never paid" [business]	**no hold or move back in the current spell**	Owner: an EPD never pays after crossing bucket 2, it keeps falling and leaves the spine	`0 0 2 3 4 4 4 4 4 4` (paying monthly) is stabilized, not EPD; `0 0 2 3 4 4 5 6` (paid once) is an early-life straight roller
Straight-roller run-up (`straight_max_runup`)	**≤ 3 months**	Any single paid month on the way down makes it a slow roller	≤ 4 moves ~70% of slow rollers into straight roller. 41% of slow rollers rest on one hold — E23 / E28 test whether that month is real
Stabilized band (`stabilized_min_in_band`) [business]	**5 of last 6 months within ±1**	The bank's existing stabilized rule, unchanged	4 of 6: share ~14%, next-month roll ~46%. 6 of 6: ~5%, ~32%
Stabilized long-run clause (`stabilized_long_run_months` , `_max_rise`) [business]	**≥ 6 months since first reaching 4, at most +1 bucket over 6 months**	Fixes oscillating payers and accounts that just paid down (§6.15). Straight rollers cannot meet it (they leave the spine within ~4 months)	Adds ~1% of the spine to stabilized. +2 would take in slow drifters paying ~4 of 6 dues; `None` switches it off

Threshold (`parameter`)	Value	Why this value	What moving it does *(synthetic)*
Repeat-delinquent depth (`repeat_min_prior_peak`) [business]	**earlier spell reached 3**	Cures from 4+ are rare; spells reaching 3 give 83% one-spell vs the bank's ~85%	From 2: more; from 4: ~5%
Repeat-delinquent window (`repeat_window_months`) [business]	**ended ≤ 12 months before this spell**	Owner decision (changed from 24); the common re-default window. Measured to the start of this spell, so the state stays fixed	Lowers the overlap with credibility's lifetime clean point (eta² 0.81 with no window → 0.52 at 12 months)
Seasoned (`seasoned_min_months`) [business]	**≥ 12 months**	The first year on book is the standard early-life window; EPD already takes ≤ 3 months, so early life = 4–11 months	Early life 4.7% at 12 m; it depends on the book's tenure mix, so the real share decides (E16)

Rule order. Q0 and EPD come before stabilized; since an EPD has never paid in its spell, an account cannot be both, so the order no longer matters for them.

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6. What was tried and dropped — and why

6.1 Unsupervised clustering as the source of states (k-means, GMM)

- **Whole spine, k-means:** very stable (bootstrap ARI 0.99) but only ARI 0.3–0.4 against the planted patterns and 0.1–0.3 against the rule states. It always merged holders with staircases and never isolated slow slides. **Stability does not mean validity** — on real data stability is the only check available, and it cannot detect merged behaviours.
- **GMM:** BIC never reached a minimum (kept improving to 12 components) because the features are spiky — 56–93% of accounts sit on a single value (run-up = 3, no earlier spell, …), and GMM fits narrow components to the spikes. Under 1% of accounts were ambiguous, so the soft-assignment advantage disappears. Not used. GMM fits continuous inputs (ability scores, balances) far better than DPD-derived counts.
- **Residual clustering (after EPD and stabilized) — a diagnostic only.** k-means at k = 6 on the rest found six nameable groups (long-clean straight roll, short-clean straight roll, earlier spell to 3+, paying after 4 but still rolling, held on the way, paid 2+ on the way), and every B/C split maps to one of them with a one-line rule. Its agreement score (ARI) is **not** evidence for B over A: the independent review showed that any finer partition built on the clustered features gains ARI — a meaningless split at median tenure gained 0.16. E22 is therefore INFO only.

6.2 A "mixed" / residual state

Dropped. An ordered tree with a default branch is MECE without one, and a residual bucket invites "everything we don't understand" to grow. (Q0 is a data-quality guard, not a behaviour bucket.)

6.3 Staircase (holds for months at each level, then steps up)

Merged. While it is holding a bucket it *is* stabilized (that is the actionable fact); just after a step it reads as a slow roller. Telling a genuine holder from a staircase mid-hold needs more than 6 months of history and an

extra rule, and the next-month risk was the same (*synthetic:* 0.28 vs 0.29).

6.4 "Paying since bucket 4" — pays after reaching 4 but still rolling (Q3)

Tested at 0.34 / 0.5 / 0.67 months paid per month since 4. Rejected:

- barely separates risk (next-month roll ~80% vs ~86%);
- unstable — ~37% leave it every month — and it drags the stability of the fixed states down by ~7pp, because accounts flip in and out of it;
- 79% of it is "paid in the last 3 months" — the credibility axis's 2-point component.

It is one of the natural residual clusters, so it is real behaviour — but it belongs to credibility. The switch is kept (`paying_since_4_min_rate`) so the test can be repeated (E24).

6.5 Separate states for slow slide, dip-and-roll, volatile

Merged into slow roller (A, B) or split once by "paid 2+ months on the way" (C). The three confuse each other in every view, and the risk actions are close. Much of "volatile" (cure then relapse) is a repeat delinquent by definition.

6.6 Repeat delinquent in option A

Left out of A on purpose. Repeat delinquent is built from earlier spells — the history credibility's 1-point "pre-delinquency clean" flag scores. In the self-test the states explain **52%** of that point's variance under the 12-month window (81% with no window). It is in B/C because it is only 1 of credibility's 5 points and *cured from bucket 3 recently* is evidence of catch-up capacity. E19b measures the real overlap and decides A vs B.

6.7 "Stabilized or improving" clause (today's bucket ≤ 6 months ago)

Dropped. It changed **0.0%** of assignments; the long-run clause (§6.15) covers the real cases.

6.8 Payment-based refinement of stabilized

Tested: "≤ 1 roll in 6 months AND paid 2+ months at least once" separated holders from staircases at 99% precision — but only because the synthetic generator built it that way, and it leans on payment behaviour (credibility). Replaced by the bank's own 5-of-6 rule.

6.9 Features that drift with time since bucket 4

Dropped from assignment: whole-spell roll rate, fixed-window counts, months paid since 4 as a count. Replaced by run-up features (fixed once past 4). The remaining current-behaviour inputs (§3) are deliberate and confined to stabilized and EPD.

6.10 Shape table as the assignment

Kept for discovery and stakeholder examples only. ~5–6k distinct banded shapes; the top 30 cover ~35–40% of accounts — too fragmented to assign everyone.

6.11 n-gram ranking by planted-label purity

Fixed. Lift is measured against each account's own codes in shuffled order: lift > 1 means the order of moves matters. No planted labels are needed.

6.12 Treating +2 jumps as impossible — and the first repair

Replaced by repair. The first version of the repair looked ahead (it used next month's digit) and also treated clean months as holds, so a "fixed" straight roller could become a slow roller when the next month arrived (`…0224` → `…02244`), and `000034` borrowed a clean month and became EPD. The repair is now causal and only uses holds at bucket 2+ (§2).

6.13 Payment-promise or inflow inputs in any state

Never used — credibility and ability own them. Their overlap is *measured* (§7).

6.14 EPD ("bust-out") with up to 6 months of history

The first version allowed ≤ 6 months, chosen to match "a few months". It made `0 0 0 0 0 0 2 3 4` an EPD — but six clean months means about four bills were paid: an **early-life default**. Its synthetic 6% match was circular: the generator's bust-out volume had been tuned to hit 6% under that rule. Replaced by the agreed definition (≤ 3 months, one blip) and, after review, the owner's "never paid since crossing 2".

6.15 Stabilized measured only around today's bucket (edge cases)

The bank's rule — 5 of the last 6 months within ±1 of **today's** bucket — misreads two uncommon but real histories (oldest → newest, today is the last digit):

- **Oscillating payer** `0000 0000 2344 4334 5433 4565 54` — swings between 3 and 6 for over a year, paying roughly what is due on average. The rule flipped it between stabilized and "straight roller" **5 times in 16 months**.
- **Just paid down** `… 3 4 5 6 6 5 4` — paid two months twice (6 → 5 → 4). The band around today's 4 excludes the two 6s, so the rule called it a straight roller *because* it paid.

Fix adopted: the **long-run clause** (§5). Both cases are now stabilized from their sixth month after first reaching 4; the only remaining dip is the single month at the top of a 3-month roll (3 → 4 → 5 → 6). Rejected alternatives: allowing +2 over 6 months (takes in slow drifters), and a "sticky" rule that remembers last month's state (harder to audit). Measured *(synthetic, with oscillating payers at 1.2% of the spine)*: of accounts ever stabilized over 6 months, the share that **left and came back** is ~5% with the clause vs ~5.5–8% on the bank's rule alone. E26 tracks this on real data; both examples are permanent regression cases.

6.16 Seasoned / early-life split in option A

Not in A: in synthetic data it adds no risk separation (by construction). It is in B/C for its action difference (life event vs underwriting signal), with the merge rule in §4.

6.17 EPD that has started paying

Before the review, an account that entered like an EPD but then paid its monthly due for 6–10 months (`0 0 2 3 4 4 4 4 4 4`) stayed EPD, with the action "early legal; no restructuring". The owner's rule — an EPD never pays after crossing bucket 2 — now sends it to stabilized.

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7. Overlap with the other axes (known and measured)

Overlap	Size *(synthetic)*	Treatment
stabilized ↔ credibility "money coming in" (2 pts)	98% of stabilized have money in; ~44% of the rest do too	One-directional: stabilized is stricter (enough to hold the bucket, 6 months). Accept and state it. E21
slow roller ↔ credibility "money coming in"	near bucket 4 almost all slow rollers paid in the last 3 months (they paid on the way down); by bucket 7 far fewer	A clock effect of the run-up, not a separate signal. Reported by state × bucket in every run
stabilized ↔ credibility "promises kept" (2 pts)	likely (paying monthly) — not testable without the score	E19 measures the total
repeat delinquent ↔ credibility "pre-delinquency clean" (1 pt)	eta² 0.52 of the clean point explained by B states (12-month window)	Decides A vs B. E19b
straight / slow roller ↔ ability	none in synthetic (by construction)	Straight rolling may proxy income loss on real data — fine, but say so. E20
any state ↔ locatability	none expected	E20b

8. Real-data validation

Two ways to run

- **File mode — for sign-off, no Spark.** `python docs/deterioration_states/run.py validate` reads `INPUT_FILE` (`data/input/dpd_synthetic_data.xlsx`, column `dpd_history_str`, CC 60–180 DPD, Aug 2026). A snapshot of today's segment has **no outcomes**: accounts that rolled to 8+ since are not in it, so roll rates and exits are SKIPPED. What it can show: shares, data quality, shapes, overlaps (if axis columns are in the file), and stability among the survivors (E13, E14, E26 run on accounts in the spine last month and today).
- **Spark mode — production.** `snippets/dpd_states_validate.py`, `main(catalog)`.

How the backtest works from a single table (Spark mode)

Today's string holds the past: dropping its newest *k* digits gives the string as it was *k* months ago, and the dropped digits are what happened next. Assumptions, each checked:

1. The table **keeps accounts that have since rolled to 8+ or closed** (a survivor-only table biases outcomes — the snippet warns when no backtest account is at 8+ today).
2. **Every string ends in the same (latest) month.** Strings frozen at charge-off or closure are not "today" — set `COL_STRING_AS_OF` to exclude them. Even 20% frozen exits bias bucket-4 roll by ~5pp while E03 still passes *(review)*.
3. **Exit codes are digits** — a write-off letter must be mapped in `CHAR_MAP`, or those accounts silently drop out (E02 now covers the backtest and the whole table).
4. The bank's roll rates (E03) are compared like for like (count- vs balance-weighted).

Outcome comparisons are **standardised to the bucket mix** (stabilized sits mostly at bucket 4) and use exit within **3 months**, because 6-month exit saturates near 100% at buckets 5–7.

Expectations (checked automatically — `dpd_states/expectations.py`)

Anchor: **bank** = the bank's figures or rules · **design** = follows from the rules · **synthetic** = calibrated on the generator only (a WARN is a prompt to look) · **data**.

	Check	Expected	Anchor
E01	Exactly one known state per account	True (FAIL)	design
E02	Unparseable strings (spine, backtest, whole table)	≤ 2% (FAIL)	data
E03	Backtest monthly roll rates vs 70/80/90/95%	≤ 7pp (FAIL)	bank
E04	Histories that start delinquent	≤ 1%	bank
E05	Unrepaired +2 jumps	≤ 2%	synthetic
E06	Bucket mix declines 4 → 7	yes	bank
E07	INFO: one-spell share, four definitions vs ~85%	report	bank
E08 / E09	EPD / stabilized share	3–8% / 4–10%	bank
E10 / E11	Recall of the bank's existing flags	≥ 80% / ≥ 90%	bank
E12	Stabilized next-month roll below the rest (bucket-standardised)	≥ 20pp	synthetic
E13	Fixed states keep their state (or move into stabilized)	≥ 95%	design
E14	Stabilized persistence next month	45–85%	synthetic
E15	Option A straight-roller share	55–80%	bank
E16	Smallest optional option-B state (EPD and stabilized are required at any size; empty = 0)	≥ 5%	design
E17	Smaller slow-roller split (C)	≥ 4%	design
E18	Stabilized at bucket 4 vs 7	≥ 2×	bank
E19 / E19b / E20 / E20b	eta² of credibility / clean point / ability / locatability by state	≤ 0.25 / 0.50 / 0.10 / 0.05	design
E21	Stabilized with money coming in	≥ 90%	design
E22	INFO: residual-cluster ARI gain (diagnostic only)	report	synthetic
E23	INFO: straight vs one-hold slow roller, next-month roll	report	—
E24	"Paying since 4" stays rejected	persistence ≤ 75%	synthetic
E25	INFO: exit-within-3m spread across fixed states (bucket-std)	report	—
E26	Stabilized accounts that leave and come back	≤ 10%	synthetic
E27	Uncropped string lengths vary (not padded / fixed-length)	p95 − p05 ≥ 6	data
E28	Run-up holds: largest calendar month vs its neighbours	≤ 2.5×	data

Every check carries an "if not met" action in the code and in the report; none of them says "tune a business parameter until it passes". The self-test also runs **24 edge cases** from the design discussion and the review, and fails if any changes state.

Choosing the option on real data

1. Any **FAIL** → fix config/data first; read nothing else.

2. **E19b high** (repeat delinquent ≈ credibility clean point) **and** the risk team does not want the cured-recently nuance → **A**. Otherwise **B**.
3. **E17 passes and** the risk team has distinct actions for intermittent vs erratic payers → **C**.
4. **E16 fails on early-life straight roller** → merge it into straight roller (agreed).
5. **E25 ≥ 5pp** → how they fell in carries risk on its own; report it to the risk team.

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9. Validation done so far *(synthetic)*

- **Calibrated generator** (`dpd_states/synth.py`): eight planted journey patterns (bust-out, clean abrupt shock, slow slide, staircase, stabilized, repeat episodes, dip-and-roll, volatile) plus a rare oscillating payer; full lifecycles to exit; length-biased snapshot like a real spine; payment holidays and billing-cycle +2 jumps; one continuous tenure distribution. Matches the bank's roll rates (71/78/89/94% vs 70/80/90/95%), single spell 83% from bucket 3 (85%) and stabilized 6.9% under the bank's rule (6%).
- **Known calibration gap:** "never paid since reaching 4" is 59% in the generator vs the owner's first estimate of ~90% (which the owner later called overkill). The generator's holders are therefore probably over-represented; this affects synthetic shares of stabilized and slow roller, not the rules.
- **Method report:** `validation/dpd_states_validation_*.md` (`run.py synthetic`).
- **Real-data path self-test:** `validation/selftest_report_*.md` (`run.py selftest`) runs the exact analysis and expectations the workplace run uses, with fake axis columns, plus the 24 edge cases. All checks pass except two expected WARNs: E16 (early-life straight roller 4.7%) and E19b (repeat delinquent ≈ clean point).
- **Independent review:** 24 findings, none critical — see REVIEW.md.

What synthetic data cannot tell you: whether the rolling states differ in outcome (they share post-4 dynamics by construction), how big the real overlaps with credibility and ability are, and how much payment holidays contaminate stabilized.

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10. Open decisions

1. Option A, B or C — decide from E19b, E17 and the risk team's actions (§8).
2. The bank's exact spell definition — E07 reports four; it matters for the 85% check and for the repeat-delinquent depth.
3. EPD vs the bank's existing bust-out rule — ours is stricter ("never paid since crossing 2"); E10 shows the agreement.
4. Restructured accounts re-opened at their old bucket (Q0) — deferred by the owner.
5. Payment holidays / restructurings — account-level flags only show how often stabilized accounts carry them (E21, section 4 of the report); month-level exclusion needs month-level flags.
6. The long-run clause widens the bank's stabilized definition (~+1% of the spine) — confirm with the owners of the existing rule (E11).

Glossary

Bucket — DPD band, one digit per month · Roll — +1 bucket, nothing paid · Hold — same bucket, one month's due paid · Cure — back to 0/1 · Spell — continuous months at ≥ 2 · Blip — 1–2 month spell never past 2 · Run-up — last clean month to first month at 4 · Spine — accounts currently at 4–7 · eta² — share of

an axis score's variance explained by the states (0 = orthogonal) · ARI — agreement between two groupings (0 = chance, 1 = identical).